SensiSpace

Feasibility Analysis &

Scalability Considerations

Satellite Land Cover Classification System

Baby Dragon Hatchling vs ResNet-18

1. Feasibility Analysis

Feasibility analysis evaluates whether the project can be successfully completed given the available resources, technology, budget, and timeline.

1.1 Technical Feasibility

Can this system be built with available technology? YES.

Aspect	Assessment
Hardware	Trained on MacBook Air (M-series). No GPU cluster needed.
Software	Open-source: Python, PyTorch, FastAPI, OpenCV.
Dataset	EuroSAT: 27,000 images, publicly free (~90MB).
Training	BDH: ~61 min, ResNet: ~4 min on single machine.
Deployment	FastAPI local server. Dockerizable for cloud.
Accuracy	95.87% - exceeds published EuroSAT benchmarks.

Technical Risks & Mitigations

Risk	Likelihood	Mitigation
GPU unavailable	Medium	BDH trains on CPU/MPS in ~1 hour
Class imbalance	Low	EuroSAT is balanced (~2,700/class)
Model overfitting	Low	Dropout + augmentation + small model
Browser compat.	Low	Standard HTML/CSS/JS, no frameworks

1.2 Economic Feasibility

Is the project cost-effective? YES. Total cost: Rs 0.

Cost Item	Amount	Notes
Hardware	Rs 0	Existing laptop (8GB+ RAM sufficient)
Software	Rs 0	Python, PyTorch, FastAPI - all open-source
Dataset	Rs 0	EuroSAT freely available from ESA
Cloud (optional)	Rs 500-1000/mo	Free tier available on AWS/GCP
Development	40 days	Single developer
TOTAL	**Rs 0-1000**	**Near-zero cost project**

Cost-Benefit Analysis

- **Manual analysis:** 1 analyst processes ~50 images/day at Rs 500/day = Rs 10/image
- **SensiSpace:** ~1,666 images/second at near-zero marginal cost
- **Break-even:** After just 100 images, the automated system saves money
- **ROI:** Effectively infinite - train once, run indefinitely

1.3 Operational Feasibility

Can end-users actually use this system? YES.

Factor	Assessment
User interface	Web-based UI, no installation. Upload + click.
Training needed	Minimal - basic computer literacy sufficient.
Response time	< 2 seconds for full classification + visualization.
Availability	Runs on any machine with Python. 24/7 capable.
Maintenance	Low - no external API dependencies.
Interpretability	Color-coded maps with labels and confidence %.

1.4 Schedule Feasibility

Can it be completed in the given timeframe? YES (Completed).

Phase	Planned	Actual	Status
Phase 1: Research & Design	16 days	16 days	Done
Phase 2: Development & Deploy	27 days	27 days	Done
Phase 3: Testing & Docs	16 days	In progress	On track
TOTAL	**59 days**	**59 days**	**On schedule**

2. Sustainability Considerations

2.1 Environmental Sustainability

Factor	Assessment
Energy use	BDH: 2.03M params. Training ~0.1 kWh (a phone charge).
Carbon footprint	Negligible. Inference at 0.6ms has near-zero cost.
Impact	POSITIVE - enables deforestation/flood monitoring.
Hardware	Consumer laptop. No specialized GPUs required.

2.2 Data Sustainability

Factor	Assessment
Availability	Sentinel-2 data free until at least 2030 (ESA).
Freshness	Retrain on newer imagery without architecture changes.
Privacy	Public satellite imagery. No personal data concerns.
Storage	Dataset ~90MB. Models <10MB each. Minimal.

2.3 Technical Sustainability

Factor	Assessment
Dependencies	Stable libs: PyTorch (Meta), FastAPI. Long-term support.
Maintainability	Modular code. Each BDH module is a separate class.
Knowledge transfer	Documented code with docstrings. Published principles.
Obsolescence risk	Low. CNNs remain standard. PyTorch actively developed.

3. Scalability Considerations

3.1 Data Scalability

Dimension	Current	Scalable To	How
Image count	27,000	Millions	Batch DataLoader, streaming
Resolution	64x64	256x256+	Adaptive pooling layers
Classes	10	50+	Change final FC layer
Spectral bands	3 (RGB)	13 (Sentinel-2)	Change stem input channels

3.2 Compute Scalability

Dimension	Current	Scalable To	Method
Single image	0.6ms	Already optimal	-
Batch processing	Sequential	1000+/batch	GPU batch inference
Multi-GPU	Single device	GPU cluster	DistributedDataParallel
Concurrent users	1	100+	Gunicorn + Nginx

3.3 Architecture Scalability (BDH)

The BDH architecture is designed to scale by adding stages, increasing channels, or adding more refinement iterations:

```
Current:   3 stages (64->128->256)           = 2.03M params
Scaled:    4 stages (64->128->256->512)       = ~5M params
Large:     5 stages (64->128->256->512->1024)  = ~15M params
```

Scaling Method	What Changes	Expected Impact
Add stages	More hierarchy levels	Better on complex scenes
Increase channels	Wider features	More land cover subtypes
More refinement	2->4 iterations	Cleaner features
Larger kernels	Add 9x9, 11x11	Better for high-res imagery

3.4 Deployment Scalability

Option	Effort	Capacity
Local (current)	Done	1 user, ~1,666 images/sec
Docker container	1 day	Portable, any cloud provider
Cloud (AWS/GCP)	3 days	Auto-scaling, 100+ concurrent users
Edge device	5 days	On-satellite or drone-mounted processing
API service	2 days	RESTful API for GIS integration

4. Future Scalability Roadmap

Timeline	Enhancement	Impact
Near-term	Higher-res imagery (256x256)	Better mixed-area classification
Near-term	Per-class confusion matrix	Identify weak classes
Mid-term	Semantic segmentation (U-Net+BDH)	True pixel-level classification
Mid-term	Multi-spectral (13 bands)	Major accuracy improvement
Long-term	Real-time satellite feed	Continuous Earth monitoring
Long-term	ISRO satellite integration	Indian-specific land analysis
Long-term	Federated learning	Privacy-preserving distributed ML

5. Summary

Dimension	Key Finding	Rating
Technical feasibility	All tools available, model works	HIGH
Economic feasibility	Near-zero cost, massive ROI	HIGH
Operational feasibility	Simple UI, minimal training	HIGH
Schedule feasibility	Completed on time	HIGH
Environmental sustainability	Low energy, positive impact	HIGH
Data sustainability	Free, continuously available	HIGH
Technical sustainability	Stable deps, modular code	HIGH
Scalability	Designed for growth	HIGH

Conclusion: The SensiSpace system is fully feasible with current technology and resources, environmentally sustainable due to its lightweight architecture (2.03M parameters, ~0.1 kWh training), and architecturally designed to scale from a single laptop to a cloud-deployed Earth monitoring service.